2008 Mathematics

Higher – Paper 1 and Paper 2

Finalised Marking Instructions

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1. Marks must be assigned in accordance with these marking instructions. In principle, marks are awarded for what is correct, rather than marks deducted for what is wrong.

2. Award one mark for each 'bullet' point. Each error should be underlined in RED at the point in the working where it first occurs, and not at any subsequent stage of the working.

3. The working subsequent to an error must be followed through by the marker with possible full marks for the subsequent working, provided that the difficulty involved is approximately similar. Where, subsequent to an error, the working is eased, a deduction(s) of mark(s) should be made.
   This may happen where a question is divided into parts. In fact, failure to even answer an earlier section does not preclude a candidate from assuming the result of that section and obtaining full marks for a later section.

4. Correct working should be ticked (√). This is essential for later stages of the SQA procedures. Where working subsequent to an error(s) is correct and scores marks, it should be marked with a crossed tick (✗ or **X**√). In appropriate cases attention may be directed to work which is not quite correct (e.g. bad form) but which has not been penalised, by underlining with a dotted or wavy line.
   Work which is correct but inadequate to score any marks should be corrected with a double cross tick (✗✗).

5. • The total mark for each section of a question should be entered in red in the **outer** right hand margin, opposite the end of the working concerned.
   • Only the mark should be written, **not** a fraction of the possible marks.
   • These marks should correspond to those on the question paper and these instructions.

6. It is of great importance that the utmost care should be exercised in adding up the marks. Where appropriate, all summations for totals and grand totals must be carefully checked. Where a candidate has scored zero marks for any question attempted, "0" should be shown against the answer.

7. As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Accept answers arrived at by inspection or mentally where it is possible for the answer so to have been obtained. Situations where you may accept such working will normally be indicated in the marking instructions.

8. Do not penalise:
   - working subsequent to a correct answer
   - legitimate variations in numerical answers
   - correct working in the "wrong" part of a question
   - omission of units
   - bad form

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9. No piece of work should be scored through without careful checking - even where a fundamental misunderstanding is apparent early in the answer. Reference should always be made to the marking scheme - answers which are widely off-beam are unlikely to include anything of relevance but in the vast majority of cases candidates still have the opportunity of gaining the odd mark or two provided it satisfies the criteria for the mark(s).

10. If in doubt between two marks, give an intermediate mark, but without fractions. When in doubt between consecutive numbers, give the higher mark.

11. In cases of difficulty covered neither in detail nor in principle in the Instructions, attention may be directed to the assessment of particular answers by making a referal to the P.A. Please see the general instructions for P.A. referrals.

12. No marks should be deducted at this stage for careless or badly arranged work. In cases where the writing or arrangement is very bad, a note may be made on the upper left-hand corner of the front cover of the script.

13 Transcription errors: In general, as a consequence of a transcription error, candidates lose the opportunity of gaining either the first ic mark or the first pr mark.

14 Casual errors: In general, as a consequence of a casual error, candidates lose the opportunity of gaining the appropriate ic mark or pr mark.

15 **Do not write any comments on the scripts**. A **revised** summary of acceptable notation is given on page 4.

16 Throughout this paper, unless specifically mentioned, a correct answer with no working receives no credit.

Summary

Throughout the examination procedures many scripts are remarked. It is essential that markers follow common procedures:

1 **Tick** correct working.
2 Put a mark in the **outer right-hand margin to match the marks allocations on the question paper.**
3 Do **no**t write marks as fractions.
4 Put each mark **at the end** of the candidate's response to the question.
5 **Follow through** errors to see if candidates can score marks subsequent to the error.
6 Do **not** write any comments on the scripts.

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2008 Higher Mathematics
Paper 1 Section A

1	C
2	D
3	C
4	B
5	A
6	B
7	C
8	D
9	B
10	A
11	B
12	C
13	A
14	B
15	C
16	A
17	C
18	C
19	B
20	D

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1.21

QU	part	mk	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
1.21	a	6	C8,C9	NC		1	3	2	6				6		
	b	5	A21,A22			1	3	1	5					5	
	c	4	C10					4	2	2			4		

A function f is defined on the set of real numbers by $f(x) = x^3 - 3x + 2$.

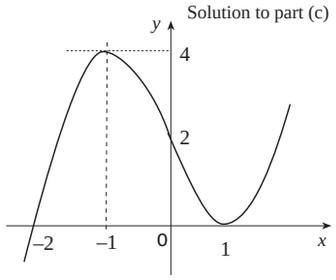
(a) Find the coordinates of the stationary points on the curve $y = f(x)$ and determine their nature. 6

(b) (i) Show that $(x-1)$ is a factor of $x^3 - 3x + 2$.

(ii) Hence or otherwise factorise $x^3 - 3x + 2$ fully. 5

(c) State the coordinates of the points where the curve with equation $y = f(x)$ meets both the axes and hence sketch the curve. 4

Solution to part (c)



The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ss	set derivative to zero
$\bullet^2$	pd	differentiate
$\bullet^3$	pd	solve
$\bullet^4$	pd	evaluate y-coordinates
$\bullet^5$	ic	justification
$\bullet^6$	ic	state conclusions
$\bullet^7$	ss	know to use $x = 1$
$\bullet^8$	pd	complete eval. & conclusion
$\bullet^9$	ic	start to find quadratic factor
$\bullet^{10}$	pd	complete quadratic factor
$\bullet^{11}$	pd	factorise completely
$\bullet^{12}$	ic	interpret y-intercept
$\bullet^{13}$	ic	interpret x-intercepts
$\bullet^{14}$	ic	sketch : showing turning points
$\bullet^{15}$	ic	sketch : showing intercepts

Primary Method : Give 1 mark for each •

$\bullet^1$	$f'(x) = 0$
$\bullet^2$	$3x^2 - 3$
	$\begin{array}{c\|c\|c\|} & \bullet^3 & \bullet^4 \\ x & -1 & 1 \\ y & 4 & 0 \end{array}$
	$\begin{array}{c\|ccc\|ccc\|} & & \bullet^5 & & & \bullet^6 & \\ & \ldots & -1 & \ldots & \ldots & 1 & \ldots \\ f' & + & 0 & - & - & 0 & + \\ & \max & at\,x = & -1 & \min & at\,x = & 1 \end{array}$
$\bullet^7$	know to use $x = 1$
$\bullet^8$	$1 - 3 + 2 = 0 \Rightarrow x - 1$ is a factor
$\bullet^9$	$(x-1)(x^2 \ldots)$
$\bullet^{10}$	$(x-1)(x^2 + x - 2)$
$\bullet^{11}$	$(x-1)(x-1)(x+2)$ ***stated explicitly***
$\bullet^{12}$	$(0, 2)$
$\bullet^{13}$	$(-2, 0)$, $(1, 0)$
$\bullet^{14}$	Sketch with turning pts marked
$\bullet^{15}$	Sketch with $(0, 2)$ or $(-2, 0)$

Notes

1 The "=0" shown at $\bullet^1$ must appear at least once before the $\bullet^3$ stage.

2 An unsimplified $\sqrt{1}$ should be penalised at the first occurrence.

3 $\bullet^3$ is only available as a consequence of solving $f'(x) = 0$.

4 The nature table must reflect previous working from $\bullet^3$.

5 Candidates who introduce an extra solution at the $\bullet^3$ stage cannot earn $\bullet^3$.

6 The use of the 2nd derivative is an acceptable strategy for $\bullet^5$.

7 As shown in the Primary Method, ($\bullet^3$ and $\bullet^4$) and ($\bullet^5$ and $\bullet^6$) can be marked in series or in parallel.

8 The working for (b) may appear in (a) or vice versa. Full marks are available wherever the working occurs.

Notes

9 In Primary method $\bullet^8$ and alternative $\bullet^9$, candidates must show some acknowledgement of the resulting "0". Although a statement wrt the zero is preferable, accept something as simple as "underlining the zero".

Alternative Method : $\bullet^7$ to $\bullet^{10}$

$\bullet^7$

1	1	0	−3	2

$\bullet^8$

1	1	0	−3	2
		1	1	−2
	1	1	−2	0

$\bullet^9$ $f(1) = 0$ *so* $(x-1)$ *is a factor*

$\bullet^{10}$ $x^2 + x - 2$

Notes

10 Evidence for $\bullet^{12}$ and $\bullet^{13}$ may not appear until the sketch.

11 $\bullet^{14}$ and $\bullet^{15}$ are only available for the graph of a cubic.

Nota Bene

For candidates who omit the x^2 coeff. leading to

$\bullet^7$ X

$\bullet^8$ $\surd$

1	1	−3	2
		1	−2
	1	−2	0

$\bullet^9$ $\surd$ $f(1) = 0$ *so* $(x-1)$.....

$\bullet^{10}$ X $x^2 - 2x$

$\bullet^{11}$ $\surd$ $x(x-1)(x-2)$

but

$\bullet^{10}$ X $x - 2$

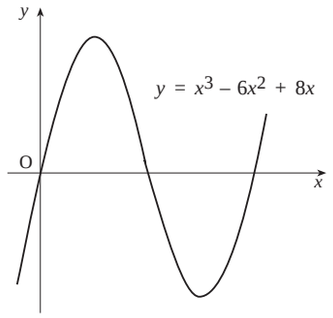
$\bullet^{11}$ X $(x-1)(x-2)$

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1.22

qu	part	mk	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
1.22	a	5	C4	NC		2	3		5				5		
	b	2	C11			1		1	2					2	

The diagram shows a sketch of the curve with equation $y = x^3 - 6x^2 + 8x$.



(a) Find the coordinates of the points on the curve where the gradient of the tangent is -1. 5

(b) The line $y = 4 - x$ is a tangent to this curve at a point A. Find the coordinates of A. 2

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ss	know to differentiate
$\bullet^2$	pd	differentiate
$\bullet^3$	ss	set derivative to -1
$\bullet^4$	pd	factorise and solve
$\bullet^5$	pd	solve for y
$\bullet^6$	ss	use gradient
$\bullet^7$	ic	interpret result

Primary Method : Give 1 mark for each $\bullet$

$\bullet^1$ $\dfrac{dy}{dx} = ..(1\ term\ correct)$ $s\ /\ i\ by\ \bullet^2$

$\bullet^2$ $3x^2 - 12x + 8$ $s\ /\ i\ by\ \bullet^3$

$\bullet^3$ $3x^2 - 12x + 8 = -1$

		$\bullet^4$	$\bullet^5$
$\bullet^4$	x	1	3
$\bullet^5$	y	3	-3

$\bullet^6$ $y = 4 - x$ has gradient $= -1$

$\bullet^7$ *check* $(3,-3)$ and reject

check $(1,3)$ and accept

Notes

1 *in* (a)

$\bullet^1$ $\surd$ $\dfrac{dy}{dx} = ..(1\ term\ correct)$

$\bullet^2$ $\surd$ $3x^2 - 12x + 8$

For candidates who now guess $x = 1$ and check that $\dfrac{dy}{dx} = -1$, only one further mark ($\bullet^3$) can be awarded. Guessing and checking further answers gains no more credit.

2 An "=0" must appear at least once in the two lines shown in the alternative for $\bullet^6$ and $\bullet^7$.

Common Error

$\bullet^1$ $\surd$ $\dfrac{dy}{dx} = ..(1\ term\ correct)$

$\bullet^2$ $\surd$ $3x^2 - 12x + 8$

$\bullet^3$ X $3x^2 - 12x + 8 = 0$

$\bullet^4$ X irrespective of what is written.

$\bullet^5$ X

Alternative for $\bullet^6$ and $\bullet^7$

$\bullet^6$
$$\begin{cases} x^3 - 6x^2 + 8x = 4 - x \\ x^3 - 6x^2 + 9x - 4 = 0 \\ (x-1)(x^2 - 5x + 4) \\ \qquad (x-4)(x-1) \end{cases}$$

$\bullet^7$ repeated root implies tangent at $(1,3)$.

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1.23

qu	part	mk	A3	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
1.23	a	3	A4	NC				3	3				3		
	b	5	A31			2	2	1		1	4				5

Functions f, g and h are defined on suitable domains by $f(x) = x^2 - x + 10$, $g(x) = 5 - x$ and $h(x) = \log_2 x$.

(a) Find expressions for $h\big(f(x)\big)$ and $h\big(g(x)\big)$. 3

(b) Hence solve $h\big(f(x)\big) - h\big(g(x)\big) = 3$ 5

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ic	interpretation composition
$\bullet^2$	ic	interpretation composition
$\bullet^3$	ic	interpretation composition
$\bullet^4$	ss	use log laws
$\bullet^5$	ss	convert to exponential form
$\bullet^6$	pd	process conversion
$\bullet^7$	pd	express in standard form
$\bullet^8$	ic	find valid solutions

Primary Method : Give 1 mark for each $\bullet$

$\bullet^1$	$h\big(f(x)\big) = h\big(x^2 - x + 10\big)$ *s / i by* $\bullet^2$
$\bullet^2$	$\log_2\big(x^2 - x + 10\big)$
$\bullet^3$	$\log_2(5 - x)$
$\bullet^4$	$\log_2\left(\dfrac{x^2 - x + 10}{5 - x}\right)$
$\bullet^5$	$\dfrac{x^2 - x + 10}{5 - x} = 2^3$
$\bullet^6$	$x^2 - x + 10 = 8(5 - x)$
$\bullet^7$	$x^2 + 7x - 30 = 0$
$\bullet^8$	$x = 3,\ -10$

Notes

1 In (a) 2 marks are available for finding one of $h(f(x))$ or $h(g(x))$ and the third mark is for the other.

2 Treat $\log_2 x^2 - x + 10$ *and* $\log_2 5 - x$ as bad form.

3 The omission of the base should not be penalised in $\bullet^2$ to $\bullet^4$.

4 $\bullet^7$ is only available for a quadratic equation and $\bullet^8$ must be the follow-through solutions.

Common Error 1

$\bullet^4$	X	$\log_2(x^2 + 5) = 3$
$\bullet^5$	$\surd$	$x^2 + 5 = 2^3$
$\bullet^6$	X	$x^2 = 3$
$\bullet^7$	X	$x = \pm\surd 3$
$\bullet^8$	X	not available

Common Error 2

$\bullet^4$	$\surd$	$\log_2\left(\dfrac{x^2 - x + 10}{5 - x}\right)$
		$\log_2\left(\dfrac{x^2 - \cancel{x} + \cancel{10}}{\cancel{5} - \cancel{x}}\right)$
		$\log_2\big(x^2 + 2\big) = 3$
$\bullet^5$	$X\surd$	$x^2 + 2 = 2^3$
$\bullet^6$	X	$x = \pm\surd 6$
$\bullet^7$	X	*not available*
$\bullet^8$	X	*not available*

Common Error 3

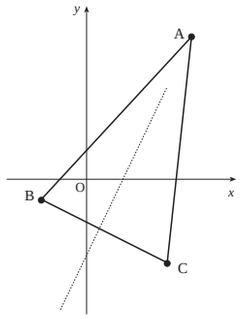
$\bullet^4$	X	not available
$\bullet^5$	$\surd$	$\log_2(x^2 - x + 10) - \log_2(5 - x) = \log_2 8$
$\bullet^6$	X	$x^2 - x + 10 - (5 - x) = 8$
$\bullet^7$	X	not available
$\bullet^8$	X	not available

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2.01

qu	part	mk	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
2.01	a	4	G7	CN		2		2	4				4		
	b	3	G7	CN		1	1	1	3				3		
	c	3	C8	CN		1	2		3				3		

The vertices of triangle ABC are A(7, 9), B(–3, –1) and C(5, –5) as shown in the diagram.
The broken line represents the perpendicular bisector of BC.



(*a*) Show that the equation of the perpendicular bisector of BC is $y = 2x - 5$. 4

(*b*) Find the equation of the median from C. 3

(c) Find the coordinates of the point of intersection of the perpendicular bisector of BC and the median from C. 3

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ss	know and find gradient
$\bullet^2$	ic	interpret perpendicular gradient
$\bullet^3$	ss	know and find midpoint
$\bullet^4$	ic	complete proof
$\bullet^5$	ss	know and find midpoint
$\bullet^6$	pd	calculate gradient
$\bullet^7$	ic	state equation
$\bullet^8$	ss	start to solve sim. equations
$\bullet^9$	pd	find one variable
$\bullet^{10}$	pd	find other variable

Primary Method : Give 1 mark for each •

$\bullet^1$	$m_{BC} = -\frac{1}{2}$	***stated explicitly***
$\bullet^2$	$m_{\perp} = 2$	*stated / implied by* $\bullet^4$
$\bullet^3$	midpoint of BC $= (1, -3)$	
$\bullet^4$	$y + 3 = 2(x - 1)$ and complete	
$\bullet^5$	midpoint of AB $= (2,4)$	
$\bullet^6$	$m_{median} = -3$	
$\bullet^7$	$y + 5 = -3(x - 5)$ *or* $y - 4 = -3(x - 2)$	
$\bullet^8$	use $y = 2x - 5$ $y = -3x + 10$	
$\bullet^9$	$x = 3$	
$\bullet^{10}$	$y = 1$	

Notes

In (a)

1 $\bullet^4$ is only available as a consequence of attempting to find and use both a perpendicular gradient and a midpoint.

2 To gain $\bullet^4$ some evidence of completion needs to be shown.
The minimum requirements for this evidence is as shown:
$y + 3 = 2(x - 1)$
$y + 3 = 2x - 2$
$y = 2x - 5$

3 $\bullet^4$ is only available for completion to $y = 2x - 5$ and nothing else.

4 Alternative for $\bullet^4$:
$\bullet^4$ may be obtained by using $y = mx + c$

Notes

In (b)

5 $\bullet^7$ is only available as a consequence of finding the gradient via a midpoint.

6 For candidates who find the equation of the perpendicular bisector of AB, only $\bullet^5$ is available.

In (c)

7 $\bullet^8$ is a strategy mark for juxtaposing the two correctly rearranged equations.

Follow - throughs

Note that from an incorrect equation in (b), full marks are still available in (c). Please follow-through carefully.

Cave

Candidates who find the median, angle bisector or altitude need to show the triangle is isosceles to gain full marks in (a).
For those candidates who do not justify the isosceles triangle, marks may be allocated as shown below:

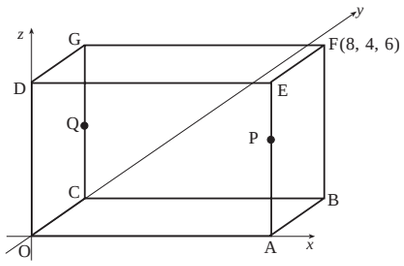
	Altitude	Median
$\bullet^1$	$\surd$	$\surd$
$\bullet^2$	$\surd$	X
$\bullet^3$	X	$\surd$
$\bullet^4$	X	X

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2.02

qu	part	mk	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
2.02	a	2	G25	CN	8202			2	2						2
	b	2	G25	CN			1	1	2						2
	c	5	G28	CR		1	4		5						5

The diagram shows a cuboid OABC,DEFG.
F is the point (8, 4, 6).
P divides AE in the ratio 2:1.
Q is the midpoint of CG.



(*a*) State the coordinates of P and Q. 2

(*b*) Write down the components of $\overrightarrow{PQ}$ and $\overrightarrow{PA}$. 2

(*c*) Find the size of angle QPA. 5

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ic	interpret ratio
$\bullet^2$	ic	interpret ratio
$\bullet^3$	pd	process vectors
$\bullet^4$	ic	interpret diagram
$\bullet^5$	ss	know to use scalar product
$\bullet^6$	pd	find scalar product
$\bullet^7$	pd	find magnitude of vector
$\bullet^8$	pd	find magnitude of vector
$\bullet^9$	pd	evaluate angle

Primary Method : Give 1 mark for each $\bullet$

$\bullet^1$ $\quad P = (8, 0, 4)$

$\bullet^2$ $\quad Q = (0, 4, 3)$

$\bullet^3$ $\quad \overrightarrow{PQ} = \begin{pmatrix} -8 \\ 4 \\ -1 \end{pmatrix}$

$\bullet^4$ $\quad \overrightarrow{PA} = \begin{pmatrix} 0 \\ 0 \\ -4 \end{pmatrix}$

$\bullet^5$ $\quad \cos QPA = \dfrac{\overrightarrow{PQ}.\overrightarrow{PA}}{|\overrightarrow{PQ}||\overrightarrow{PA}|}$ *stated / implied by* $\bullet^9$

$\bullet^6$ $\quad \overrightarrow{PQ}.\overrightarrow{PA} = 4$

$\bullet^7$ $\quad |\overrightarrow{PQ}| = \sqrt{81}$

$\bullet^8$ $\quad |\overrightarrow{PA}| = \sqrt{16}$

$\bullet^9$ $\quad 83{\cdot}6°,\ 1.459$ *radians*, 92.9 *gradians*

Notes

1 Treat coordinates written as column vectors as bad form.

2 Treat column vectors written as coordinates as bad form.

3 For candidates who do not attempt $\bullet^9$, the formula quoted at $\bullet^5$ must relate to the labelling in order for $\bullet^5$ to be awarded.

4 Candidates who evaluate $P\hat{O}Q$ correctly gain 4/5 marks in (c) (74° or 75°)

Exemplar 1

$\bullet^3$,$\bullet^4$ X, X $\quad \overrightarrow{OA} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}$ $\quad \overrightarrow{OQ} = \begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix}$

$\bullet^5$ $\quad X$ $\quad \cos AOQ = \dfrac{\overrightarrow{OA}.\overrightarrow{OQ}}{|\overrightarrow{OA}||\overrightarrow{OQ}|}$

$\bullet^6$ $\quad \surd$ $\quad \overrightarrow{OA}.\overrightarrow{OQ} = 0$

$\bullet^7$ $\quad \surd$ $\quad |\overrightarrow{OA}| = \sqrt{64}$

$\bullet^8$ $\quad \surd$ $\quad |\overrightarrow{OQ}| = \sqrt{25}$

$\bullet^9$ $\quad \surd$ $\quad 90°$

Exemplar 2

$\bullet^3$,$\bullet^4$ X, X $\quad \overrightarrow{OA} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}$ $\quad \overrightarrow{OQ} = \begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix}$

$\bullet^6$ $\quad \surd$ $\quad \overrightarrow{OA}.\overrightarrow{OQ} = 0$

$\bullet^9$ $\quad \surd$ $\quad 90°$

Alternative for $\bullet^5$ to $\bullet^8$

$\bullet^5$ $\quad \cos QPA = \dfrac{PA^2+PQ^2-QA^2}{2PA \times PQ}$

$\bullet^6$ $\quad |\overrightarrow{PA}| = \sqrt{16}$

$\bullet^7$ $\quad |\overrightarrow{PQ}| = \sqrt{81}$

$\bullet^8$ $\quad |\overrightarrow{QA}| = \sqrt{89}$

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2.03

qu	part	2	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
2.03	a	2	T4	CN	8203			2	2				2		
	b	4	T13	CR		1	2	1	4						4
	c	2	C20	CN			1	1	1	1					2

(*a*) (i) Diagram 1 shows part of the graph of $y = f(x)$, where $f(x) = p\cos x$.
Write down the value of p.

(ii) Diagram 2 shows part of the graph of $y = g(x)$, where $g(x) = q\sin x$.
Write down the value of q. 2

(*b*) Write $f(x) + g(x)$ in the form $k\cos(x + a)$ where $k > 0$ and $0 < a < \frac{\pi}{2}$. 4

(*c*) Hence find $f'(x) + g'(x)$ as a single trigonometric expression. 2

Diagram 1

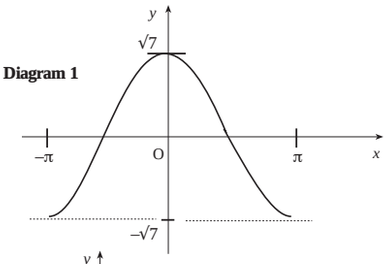
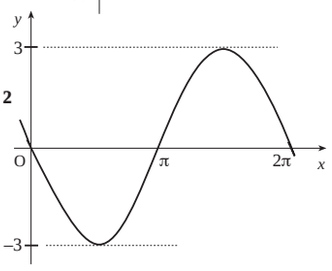


Diagram 2



The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

•1	ic	interpret graph
•2	ic	interpret graph
•3	ss	expand
•4	ic	compare coefficients
•5	pd	process "k"
•6	pd	process "a"
•7	ss	state equation
•8	pd	differentiate

Primary Method : Give 1 mark for each •

•1	$p = \sqrt{7}$
•2	$q = -3$
•3	$k\cos x\cos a - k\sin x\sin a$ ***stated explicitly***
•4	$k\cos a = \sqrt{7}$ ***and*** $k\sin a = 3$ ***stated explicitly***
•5	$k = 4$
•6	$a \approx 0{\cdot}848$
•7	$4\cos(x + 0{\cdot}848)$
•8	$-4\sin(x + 0{\cdot}848)$

Notes

In (a)

1 For •1 accept $p = 2.6$ leading to $k = 4.0$, $a = 0.86$ in (b).

In (b)

2 $k(\cos x\cos a - \sin x\sin a)$ is acceptable for •3.

3 Treat $k\cos x\cos a - \sin x\sin a$ as bad form only if the equations at the •4 stage both contain k.

4 $4(\cos x\cos a - \sin x\sin a)$ is acceptable for •3 and •5.

5 $k = \sqrt{16}$ does not earn •5.

6 No justification is needed for •5.

7 Candidates may use any form of wave equation as long as their final answer is in the form $k\cos(x + a)$. If not, then •6 is not available.

Notes

8 Candidates who use degrees throughout this question lose •6, •7 and •8.

Common Error 1

(*sic*)

$q = 3 \quad \Rightarrow k = 4,\ \tan a = -\frac{3}{\sqrt{7}}$

$\Rightarrow a = 5.44 \ \ or\ \ -0.85$

•2 X, •3 √, •4 √, •5 √, •6 √

Common Error 2

(*sic*)

$q = 3 \quad \Rightarrow k = 4,\ \tan a = -\frac{3}{\sqrt{7}}$

$\Rightarrow a = 0.85$

•2 X, •3 √, •4 √, •5 √, •6 X

Note that •6 is not awarded as it is not consistent with previous working.

Alternative Method (for •7 and •8)

If :

$f'(x) + g'(x) = -\sqrt{7}\sin x - 3\cos x$ .....

then •7 is only available once the candidate has reached e.g.

"choose $k\sin(x + a)$

$\Rightarrow k\sin a = -3,\ k\cos a = -7$."

•8 is available for evaluating k and a.

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2.04

qu	part	mk	code	calc	source	ss		ic	C	B	A		U1	U2	U3
2.04	a	2	G9	CN	8204			2	2					2	
	b	4	G14	CN		1	1	2	2	2				4	
	c	5	G12	CN		1	4			5				5	

(*a*) Write down the centre and calculate the radius of the circle with equation $x^2 + y^2 + 8x + 4y - 38 = 0$. 2

(*b*) A second circle has equation $(x-4)^2 + (y-6)^2 = 26$.
Find the distance between the centres of these two circles and hence show that the circles intersect. 4

(*c*) The line with equation $y = 4 - x$ is a common chord passing through the points of intersection of the two circles.
Find the coordinates of the points of intersection of the two circles. 5

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ic	state centre of circle
$\bullet^2$	ic	find radius of circle
$\bullet^3$	ic	state centre and radius
$\bullet^4$	pd	find distance between centres
$\bullet^5$	ss	find sum of radii
$\bullet^6$	ic	interpret result
$\bullet^7$	ss	know to and substitute
$\bullet^8$	pd	start process
$\bullet^9$	pd	write in standard form
$\bullet^{10}$	pd	solve for x
$\bullet^{11}$	pd	solve for y

Primary Method : Give 1 mark for each $\bullet$

$\bullet^1$	$(-4,-2)$		
$\bullet^2$	$\sqrt{58}$ (≈ 7.6)		
$\bullet^3$	$(4,6)$ ***and*** $\sqrt{26}$ (≈ 5.1) *s / i* $\bullet^4$ *and* $\bullet^5$		
$\bullet^4$	$d_{centres} = \sqrt{128}$ *accept* 11.3		
$\bullet^5$	$\sqrt{58} + \sqrt{26}$ *accept* 12.7		
$\bullet^6$	compare 12.7 and 11.3		
$\bullet^7$	$x^2 + (4-x)^2 + \ldots$		
$\bullet^8$	$x^2 + 16 - 8x + x^2 + \ldots$		
$\bullet^9$	$2x^2 - 4x - 6 = 0$		
		$\bullet^{10}$	$\bullet^{11}$
$\bullet^{10}$	x	3	-1
$\bullet^{11}$	y	1	5

Notes

In (a)

1 If a linear equation is obtained at the $\bullet^9$ stage, then $\bullet^9$, $\bullet^{10}$ and $\bullet^{11}$ are not available.

2 Solving the circles simultaneously to obtain the equation of the common chord gains no marks.

3 The comment given at the $\bullet^6$ stage must be consistent with previous working.

alt. for $\bullet^7$ to $\bullet^{11}$:

$\bullet^7$	$(4-y)^2 + \ldots$		
$\bullet^8$	$y^2 - 8y + 16 + y^2 + \ldots$		
$\bullet^9$	$y^2 - 6y + 5 = 0$		
		$\bullet^{10}$	$\bullet^{11}$
$\bullet^{10}$	y	1	5
$\bullet^{11}$	x	3	-1

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2.05

qu	part	mk	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
2.05		5	T10	CR		1	4			5				5	

Solve the equation $\cos 2x^\circ + 2\sin x^\circ = \sin^2 x^\circ$ in the interval $0 \leq x < 360$. 5

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ss	use double angle formula
$\bullet^2$	pd	obtains standard form (i.e. " ..... = 0")
$\bullet^3$	pd	factorise
$\bullet^4$	pd	process factors
$\bullet^5$	pd	completes solutions

Primary Method : Give 1 mark for each $\bullet$

$\bullet^1$	$\cos 2x = 1 - 2\sin^2 x$	
$\bullet^2$	$3\sin^2 x - 2\sin x - 1 = 0$	
$\bullet^3$	$(3\sin x + 1)(\sin x - 1) = 0$	
	$\bullet^4$	$\bullet^5$
$\bullet^4$	$\sin x = -\frac{1}{3}$	$\sin x = 1$
$\bullet^5$	$199.5^\circ, 340.5^\circ$	90°

Notes

1 $\bullet^1$ is not available for $1 - 2\sin^2 A$ with no further working.

2 $\bullet^2$ is only available for the three terms shown written in any correct order.

3 The "=0" has to appear at least once "en route" to $\bullet^3$.

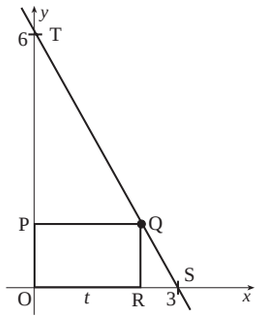
4 $\bullet^4$ and $\bullet^5$ are only available for solving a quadratic equation.

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2.06

qu	part	mk	code	calc	source	ss	pd	ic	C	B	A		U1	U2	U3
2.06		3	G3	CN	8206	1		2			3		3		
		6	C11	CN		2	2	2		6			6		

In the diagram Q lies on the line joining (0, 6) and (3, 0).
OPQR is a rectangle, where P and R lie on the axes and OR $= t$.



(a) Show that QR $= 6 - 2t$. 3

(b) Find the coordinates of Q for which the rectangle has a maximum area. 6

The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

$\bullet^1$	ss	know and use e.g. similar triangles, trigonometry or gradient
$\bullet^2$	ic	establish equation
$\bullet^3$	ic	find a length
$\bullet^4$	ss	know how and find area
$\bullet^5$	ss	set derivative of the area function to zero
$\bullet^6$	pd	differentiate
$\bullet^7$	pd	solve
$\bullet^8$	ic	justify stationary point
$\bullet^9$	ic	state coordinates

Primary Method : Give 1 mark for each $\bullet$

$\bullet^1$	ΔOST, RSQ are similar s / i by $\bullet^2$
$\bullet^2$	$\frac{\text{QR}}{6} = \frac{3-t}{3}$ or equivalent
$\bullet^3$	QR $= 6 - 2t$
$\bullet^4$	$A(t) = t(6-2t)$
$\bullet^5$	$A'(t) = 0$
$\bullet^6$	$6 - 4t$
$\bullet^7$	$t = \frac{3}{2}$
$\bullet^8$	*e.g. nature table*
$\bullet^9$	$\text{Q} = \left(\frac{3}{2}, 3\right)$

Notes

1 "$y = 6 - 2x$" appearing *ex nihilo* can be awarded neither $\bullet^1$ nor $\bullet^2$.
$\bullet^3$ is still available with some justification *e.g.* OR $= t$ gives $y = 6 - 2t$.

2 The "=0" has to appear at least once before the $\bullet^7$ stage for $\bullet^5$ to be awarded.

3 Do not penalise the use of $\frac{\text{dy}}{\text{dx}}$ in lieu of A$'(t)$ for instance in the nature table.

4 The minimum requirements for the nature table are shown on the right.
Of course other methods may be used to justify the nature of the stationary point(s).

Variation 1:

$\bullet^1$ $\tan\text{'S'} = \frac{6}{3}$

$\bullet^2$ $\tan\text{'S'} = \frac{\text{QR}}{3-t}$ *and equate*

Variation 2 :

$\bullet^1$ $\surd m_{\text{line}} = -2$ s / i *by* $\bullet^2$

$\bullet^2$ $\surd$equation of line $: y = -2x + 6$

Variation 3

$\bullet^1$ $\surd$ $m_{\text{line}} = -2$

$\bullet^2$ $\surd$equation of line $: y = 6 - 2x$

Variation 4

$\bullet^1$ X (*nothing stated*)

$\bullet^2$ Xequation of line $: y = 6 - 2x$

Alternative Method : (for $\bullet^5$ *to* $\bullet^8$)

$\bullet^5$ strategy to find roots $\Rightarrow$ t.p.s

$\bullet^6$ $t = 0$, $t = 3$

$\bullet^7$ max *t.p.* since coeff of "t^2" < 0

$\bullet^8$ turning pt at $t = \frac{3}{2}$

Nature Table

minimum requirements for $\bullet^8$

		$\frac{3}{2}$	
A'	$+$	0	$-$
$\bullet^8$	$\ddots$	$\cdots$	$\ddots$

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2.07

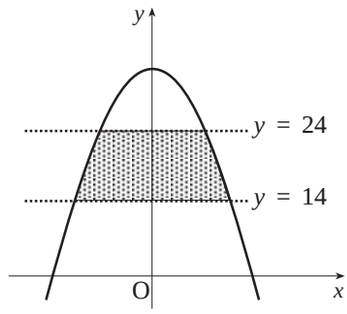
qu	part	mk	code	calc	source	ss	pd	ic	C	B	A	U1	U2	U3
2.07		8	C19	CN		3	4	1			8		8	

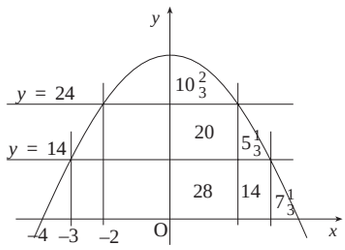
The parabola shown in the diagram has equation

$y = 32 - 2x^2$.

The shaded area lies between the lines $y = 14$ and $y = 24$.

Calculate the shaded area. 8





The primary method is based on this generic marking scheme which may be used as a guide for any method not shown in detail.

Generic Marking Scheme

•1	ic	interpret limits
•2	pd	find both x-values
•3	ss	know to integrate
•4	pd	integrate
•5	ic	state limits
•6	pd	evaluate limits
•7	ss	select "what to add to what"
•8	pd	completes a valid strategy

Primary Method : Give 1 mark for each •

•1	$32 - 2x^2 = 24$ ***or*** 14
•2	$x = 2$ ***and*** 3
•3	$\int (32 - 2x^2) \quad dx$
•4	$32x - \frac{2}{3}x^3$
•5	$\left[\ldots\right]_2^3$
•6	$19\frac{1}{3}$
•7	*e.g.* $19\frac{1}{3} - 14 + 20$ and then double s / i by •8
•8	$50\frac{2}{3}$

Notes

1 For $\int_{14}^{24} (32 - 2x^2)dx \quad = \left[32x - \frac{2}{3}x^3\right]$

may be awarded •3 and •4 *ONLY*.

2 **For integrating "along the y - axis"**

•1 strategy: choose to integrate along y-axis

•2 $x = \sqrt{\left(16 - \frac{1}{2}y\right)}$

•3 $\int \left(16 - \frac{1}{2}y\right)^{\frac{1}{2}} \quad dy$

•4 $-2.\frac{2}{3}\left(16 - \frac{1}{2}y\right)^{\frac{3}{2}}$

•5 $\left[\ldots\right]_{14}^{24}$

•6 $-\frac{4}{3}(4^{\frac{3}{2}} - 9^{\frac{3}{2}})$

•7 $2\times$.....

•8 $50\frac{2}{3}$

Exemplar 1 (•3 *to* •8)

•3 $\int (32 - 2x^2 - 14) \quad dx$

•4 $18x - \frac{2}{3}x^3$

•5 $\left[\ldots\right]_{-3}^{3}$

•6 72

•7 *e.g.* $72 - \int_{-2}^{2} (32 - 2x^2 - 24) \quad dx$

•8 $50\frac{2}{3}$

or

•5 $\left[\ldots\right]_0^3$

•6 36

•7 *e.g.* $2\times\left[36 - \int_0^2 (32 - 2x^2 - 24) \quad dx\right]$

Variations (•3 *to* •6)

The following are examples of sound opening integrals which will lead to the area after one more integral at most.

$\int_0^2 (32 - 2x^2)dx \quad = \ldots\ldots = 58\frac{2}{3}$

$\int_0^3 (32 - 2x^2)dx \quad = \ldots\ldots = 78$

$\int_2^3 (32 - 2x^2)dx \quad = \ldots\ldots = 19\frac{1}{3}$

$\int_0^2 (32 - 2x^2 - 24)dx \quad = \ldots\ldots = 10\frac{2}{3}$

$\int_0^3 (32 - 2x^2 - 14)dx \quad = \ldots\ldots = 36$

$\int_2^3 (32 - 2x^2 - 14)dx \quad = \ldots\ldots = 5\frac{1}{3}$

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